SWEETPRINT CNC SYSTEM

PROJECT

By SYNTAX ERROR (B25)

SYNTAX ERROR

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Introduction

Automation is widely used in the modern food industry to improve efficiency and consistency.

Traditional cake decoration is:

- Time-consuming
- Skill-dependent
- Inconsistent in quality

To overcome these issues, this project proposes a Cake Icing CNC Machine, which uses Computer Numerical Control (CNC) technology to automatically decorate cakes.

The system works similarly to a 3D printer, where a controlled nozzle extrudes icing to create precise designs.

AIM

To design and develop a low-cost, food-safe automated cake icing CNC machine.

Objectives

- Develop a CNC-based X-Y-Z motion system

Design a controlled icing extrusion mechanism

Implement Bluetooth (SPP) based control

Use G-code for design execution

Ensure food safety and hygiene

Achieve accurate and repeatable decoration

Problem and Proposed Solution

Problem

Manual cake decoration: Requires skilled labor

Manual cake decoration: Takes more time

Manual cake decoration: Produces inconsistent results

Industrial machines: Expensive

Industrial machines: ExNot suitable for small businesse spensive

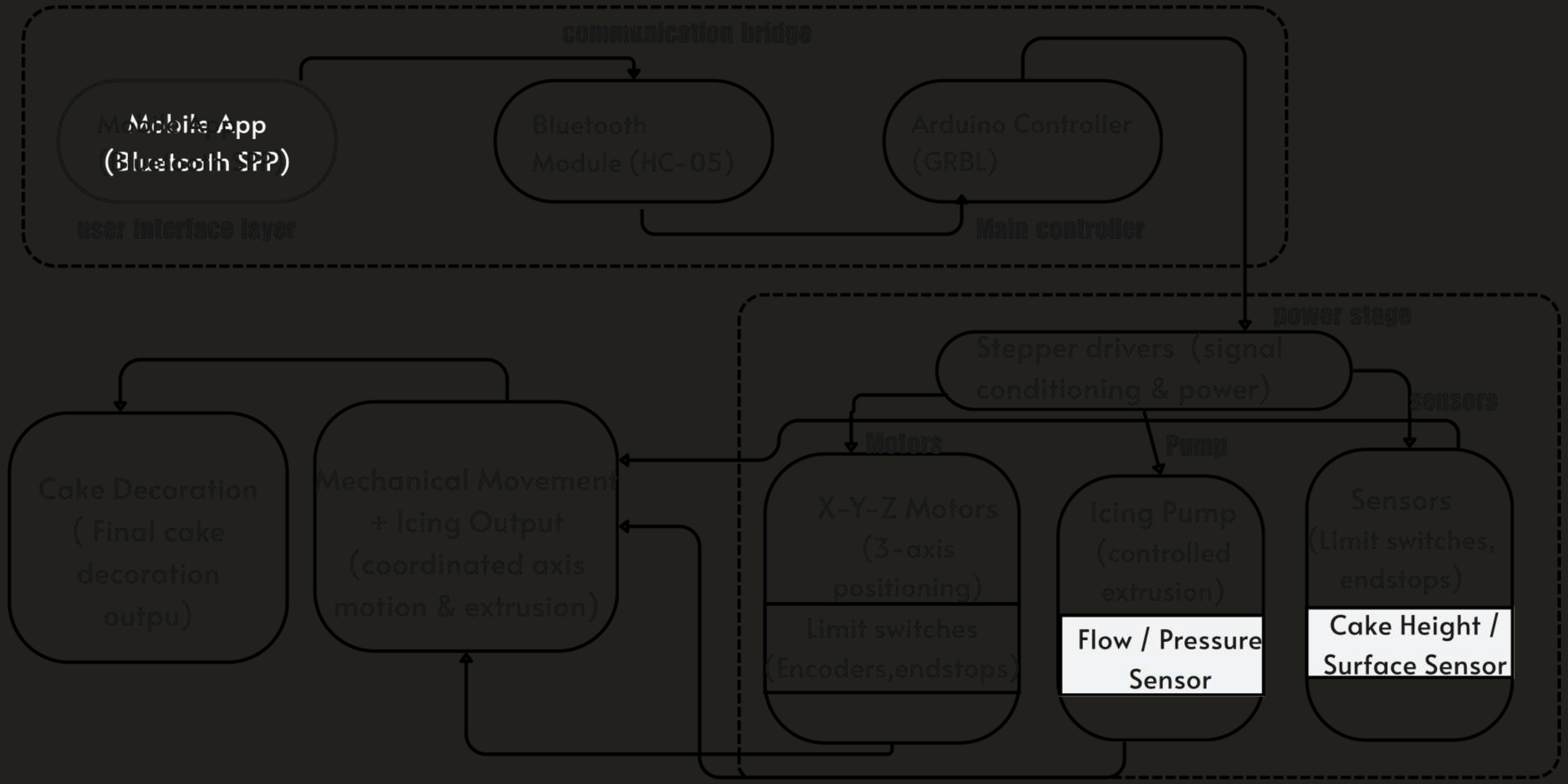
Proposed Solution

Develop a low-cost automated CNC icing machine that:

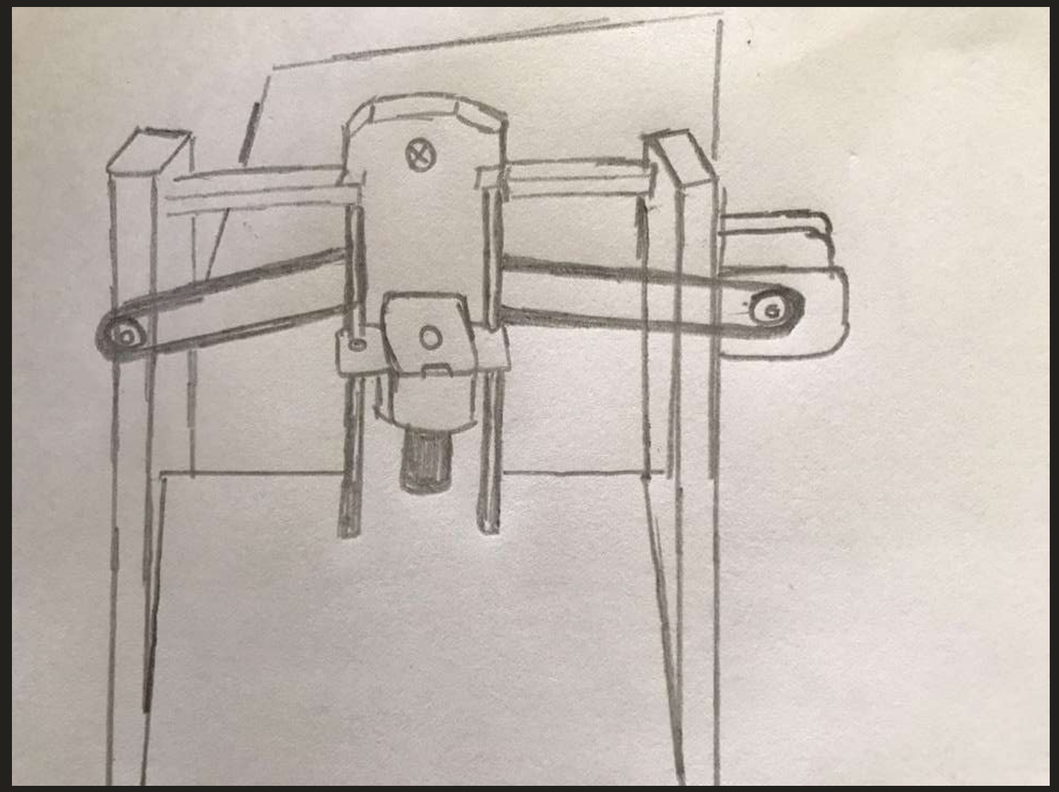
- Performs automatic cake decoration
- Maintains consistent quality
- Saves time and labor

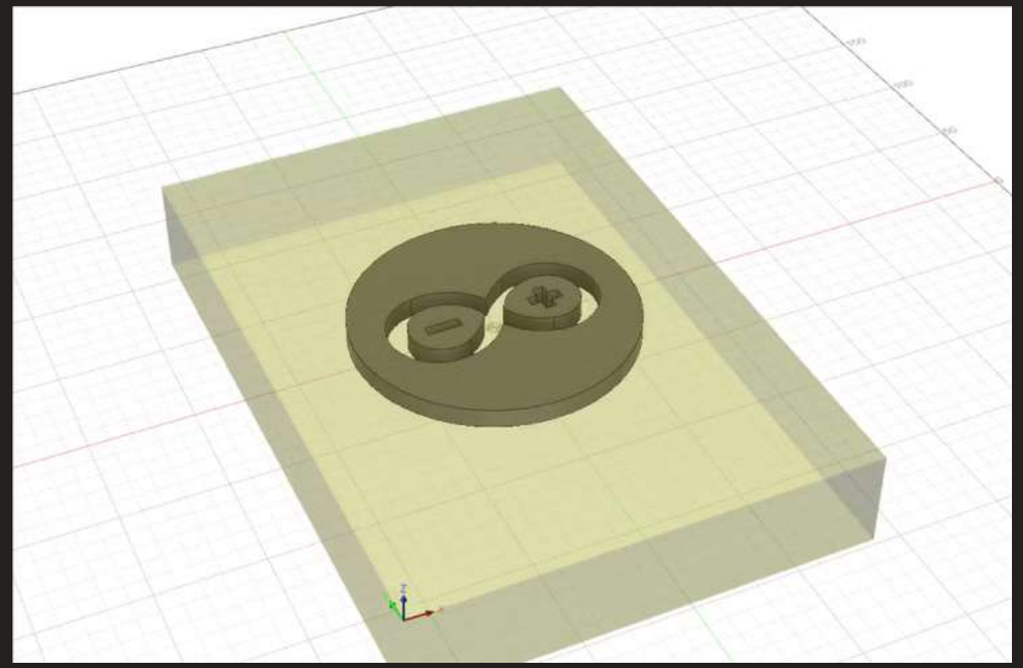
Can be controlled via mobile using Bluetooth (SPP)

Block Diagram


communication bridge
Mobile App
(Bluetooth SPP)
Bluetooth
Module (HC-05)
Arduino Controller
(GRBL)
user interface layer
Main controller
power stage
Stepper drivers (signal
conditioning & power)
sensors
Motors
Pump
Cake Decoration
(Final cake
decoration
outpu)
Mechanical Movement
+ Icing Output
(coordinated axis
motion & extrusion)
X-Y-Z Motors
(3-axis
positioning)
Limit switches
(Encoders,endstops)
Icing Pump
(controlled
extrusion)
Flow / Pressure
Sensor
Sensors
(Limit switches,
endstops)
Cake Height /
Surface Sensor

3D MODEL





FINAL OUTPUTS

Expected Budget

Component / Module	Quantity	Cost (LKR)
Arduino Uno / ESP32	1	2,500
Stepper Motors (NEMA 17)	3	9,000
Stepper Motor Drivers (A4988/DRV8825)	3	2,500
CNC Frame Materials	1	10,000
Bluetooth Module (HC-05)	1	1,500
Syringe / Icing Pump System	1	5,000
Limit switches / Encoders	7	2,000
Flow / Pressure Sensor	1	2,000
Cake Height / Surface Sensor	1	1,500
Power Supply + Voltage Regulator	1	3,000
Wiring, connectors, misc. electronics	-	2,500

Miscellaneous / Safety	-	2, 000

WORK DISTRIBUTION

Member 1 – Mechanical Design Engineer
Responsibilities:

- Design the CNC frame structure
- Develop X-Y-Z axis movement system
- Install rails, belts, and motors
- Ensure mechanical stability and accuracy

Member 2 – Icing System & Food Safety Engineer
Responsibilities:

Design icing extrusion system (syringe/pump)
Select food-grade materials (nozzle, tubes)
Implement hygienic design (removable parts)
Develop cleaning mechanism

Member 3 – Electronics Engineer
Responsibilities:

Connect Arduino and CNC shield
Interface stepper motors and drivers
Integrate power supply system
Connect Bluetooth module

Member 4 – Software & Control Engineer
Responsibilities:

- Install and configure GRBL firmware
- Generate and manage G-code
- Develop control logic for motion + icing
- Ensure synchronization of system

Member 5 – UI, Testing & Documentation Engineer
Responsibilities:

Develop mobile app (Bluetooth control)
Handle user interface (start/stop, settings)
Test system performance and accuracy
Prepare documentation and presentation

References

- Arduino Official Documentation
- GRBL Firmware Documentation
- CNC Machine Design Resources
- Inkscape G-code Tutorials
- General Food Safety Guidelines

THANK YOU